

3. In man after the infection

When an infected mosquito bites a man, the 3rd stage microfilaria larvae enter the blood circulation of man and finally reach the lymphatic vessels. Here they undergo the 3rd and the 4th moultings to produce young filarial worms. They attain sexual maturity within 5 to 18 months.

III. Pathogenicity

Light infection causes filarial fever which is characterised by headache, mental depression and increase in the body temperature. In general, the infection of filarial worm causes inflammatory effect in lymph vessels and lymph glands. Inflammation of the lymph vessels is called **lymphangitis** (Gr., *angeos* - vessels, *itis* - inflammation) and that of lymph glands is called **lymphadenitis** (Gr., *adenos* - gland, *itis* - inflammation). In the case of heavy infection, the accumulation of dead worms blocks the lymph vessels and lymph glands resulting in immense swelling. This condition is called **lymphoedema** (Gr., *oiedema* - a swelling) which is noticed in the extremities of limbs, scrotum of males and mammary glands in females. Fibroblasts accumulate in these tissues and form the fibrous tissue. In severe cases, the sweat glands of the skin in the affected regions disintegrate and the skin becomes rough. This terminal condition is referred to as '**elephantiasis**'.

IV. Prophylaxis: Same as that of *Plasmodium*.

6.3 BRIEF ACCOUNT OF SOME OTHER DISEASES

6.3.1 Bacterial diseases

- i) **Typhoid fever:** It is caused by *Salmonella typhi* which is a Gram negative bacterium. It mainly lives in the small intestine of man and then migrates to other organs through blood. It can be confirmed by **Widal test**.

Mode of infection: Contamination through food and water.

Symptoms: Sustained fever with high temperature upto 104°F, weakness, stomach pain, constipation, headache and loss of appetite. Intestinal perforation and death may also occur in severe cases.

A classic case in medicine: Mary Mallon nicknamed Typhoid Mary, was a cook by profession and was a typhoid carrier. She continued to spread this disease for several years through the food she prepared.

- ii) **Pneumonia:** It is caused by Gram positive bacteria such as ***Streptococcus pneumoniae*** and ***Haemophilus influenzae***. They infect the alveoli of lungs in human beings.

Mode of infection: Contamination by inhaling the droplets/aerosols released by an infected person or even by sharing the utensils with an infected person.

Symptoms: The alveoli get filled with fluid leading to severe problems in respiration. In severe cases, the lips and finger nails may turn gray to bluish in colour.

6.3.2 Viral Diseases

It is not an exaggeration to say that there is no person in this world who has not suffered from cold at least once in his life time.

Common cold: It is caused by **Rhino virus** group of viruses. They infect nose and respiratory passage but not lungs.

Mode of infection: Contamination by direct inhalation of the droplets resulting from cough or sneezes of an infected person or indirectly through contaminated objects such as pens, books, cups, door-knobs, computer keyboard or mouse etc.

Generally all the medicines that are used against cold cause drowsiness. Try to find out the reason for this from your family doctor.

Symptoms: Nasal congestion, discharge from nose, sore throat, hoarseness, cough, headache, tiredness, etc., which usually last for 3-7 days.

Do you know: 'Cold' treated is cured in just **ONE** week, and 'cold' untreated is cured in **SEVEN DAYS**. What is the implied meaning?

6.3.3 Fungal Diseases



Fig. 6.13 Ringworm infection

Ringworm: It is one of the most common infectious diseases in man. It is caused by many fungi belonging to the genera ***Microsporum***, ***Trichophyton*** and ***Epidermophyton***. Heat and moisture help these fungi grow in the skin folds such as those in the groin or between the toes.

Mode of infection: Contamination by using towels, clothes or combs of the infected persons or even from soil.

Symptoms: Appearance of dry, scaly, usually round lesions accompanied by intense itching on various parts of the body such as skin, nails and scalp.

6.3.4 Prophylaxis

- i) **In the case of bacterial & viral diseases:** The advancements made in biological science have armed us to deal with many infectious diseases effectively. The immunization programmes by the use of vaccines have enabled us to completely eradicate a deadly disease like smallpox. A large number of other infectious diseases like polio, diphtheria, pneumonia and tetanus have been controlled to a large extent by the use of vaccines.
- ii) **In general:** Biotechnology is at the verge of making available newer and safer vaccines. Discovery of antibiotics and various other drugs has also enabled us to treat infectious diseases effectively.